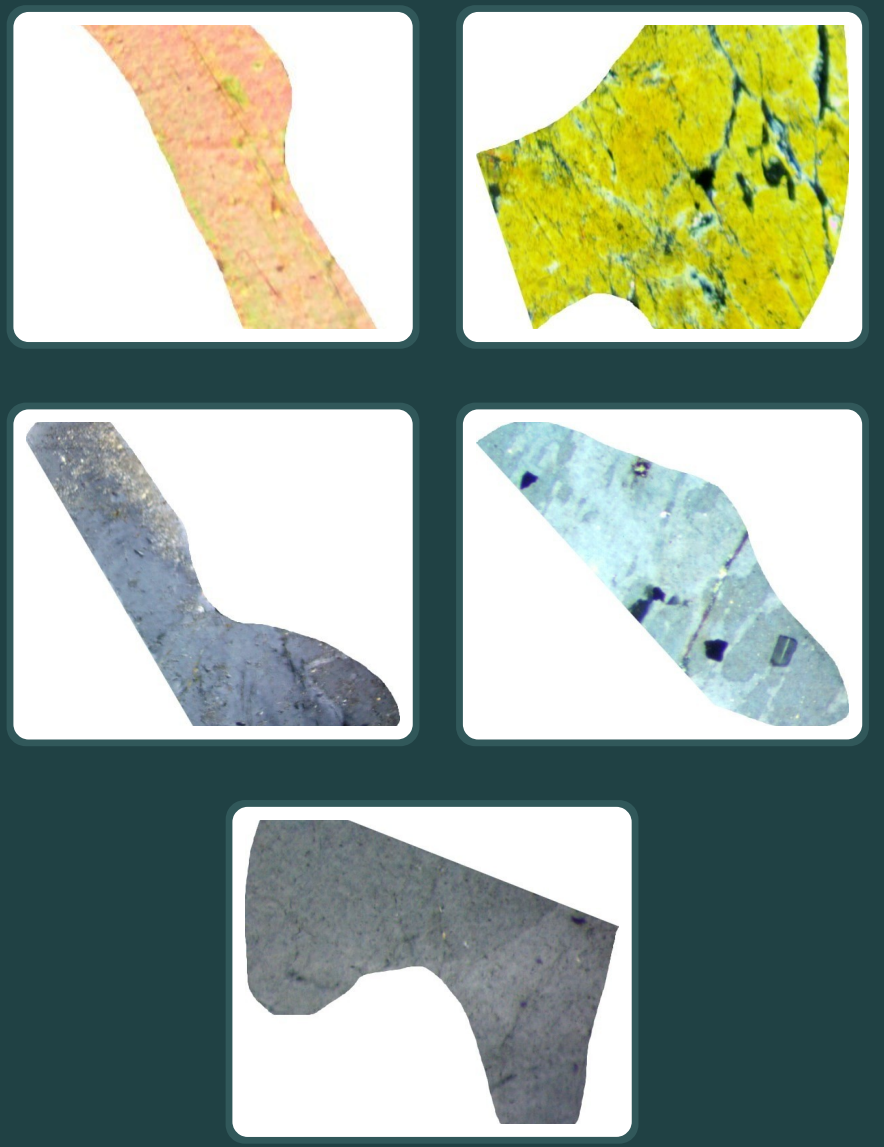
REPRODUCTION • CRITICAL EVALUATION

100% ACCURACY

— BUT ON WHAT?

Mineral photomicrograph classification through the lens of dataset quality



Reproducing Amer et al. (2025), Scientific Data

Why mineral recognition is challenging

The same mineral can look very different when the microscope stage rotates.



0°



60°



150°

ONE MINERAL

many visual appearances

Illustrative quartz images from one named acquisition folder in the experimental archive.

Visuals: MUMDMC2025 experimental archive; acquisition geometry: Amer et al. (2025).

What the paper set out to deliver

Amer et al. (2025) propose a systematic photomicrographic mineral dataset and validate it with classical ML.

14,400

images in the full collection

72 rotations × 5 minerals × 2 modes × 20 crystals

2,500

documented public XPL subset

500 photographs per mineral

87.2%

reported KNN accuracy

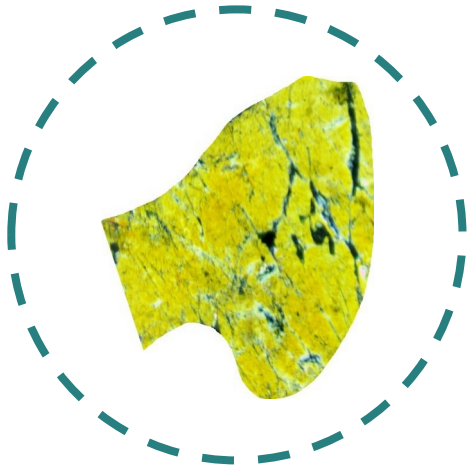
on resized 150 × 150 images

Their premise: systematic acquisition + traceable metadata should support reliable mineral classification.

Source: Amer et al., "A Photomicrographic Dataset of Rocks for the Accurate Classification of Minerals," Scientific Data (2025), Table 3.

How the data were produced

The original acquisition protocol samples rotations systematically, not independent specimens.



0° → 5° → 10° → … → 355°

72 frames / revolution

DECLARED FILENAME SCHEME

F61-Quartz-CN-1-8-35deg

F61 thin section	**Quartz** mineral	**CN** polarization
1 crystal	**8** frame	**35°** rotation

Source: Amer et al. (2025), dataset acquisition and filename convention, pp. 7–9.

The first reproducibility gap

The archive we could actually use did not match the 2,500-image subset described by the paper.

14,400

Full collection described

Research paper · PPL + XPL

2,500

Public XPL subset described

Metadata spreadsheets · 500/class

583

Images in our working ZIP

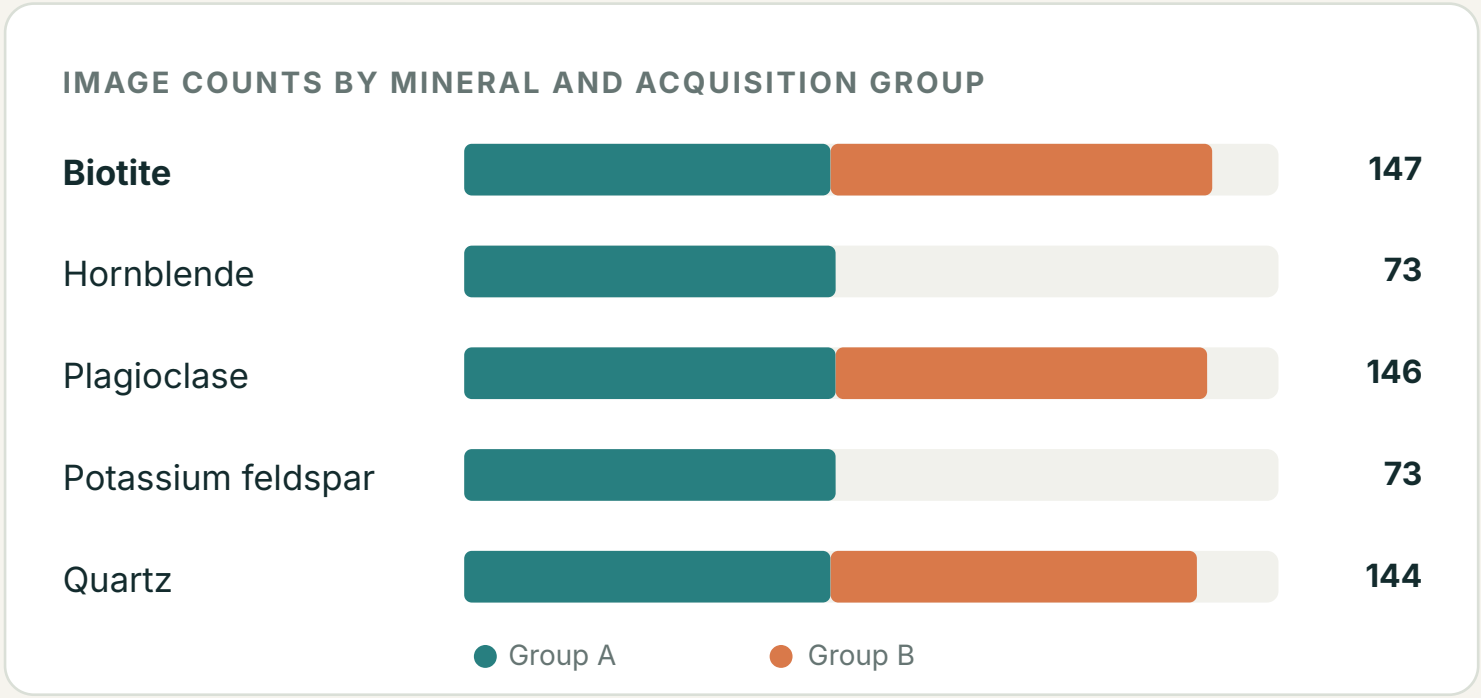
Five classes · 8 acquisition folders

Scope of our claim: a mismatch in the archive used for reproduction, not proof that every published file is unavailable.

Sources: Scientific Data (2025); accompanying five metadata spreadsheets (500 rows each); experimental ZIP count.

Eight acquisition groups, uncertain identities

The files form eight rotational folders. A folder may represent a crystal—but that is not independently verified.



176 filenames repeated across folders

Working hypothesis

8 groups ≈ 8 individual crystals

Visual inference, not validated metadata

Sources: experimental ZIP folder audit and accompanying 2,500-record metadata tables.

Reproducing the classification pipeline

We followed the article's classical-ML logic on the subset available to us.



Important implementation detail The notebook's "Original" branch standardizes variable crops to 224 × 224. It is not native microscope resolution.

Method: article, Table 3 and ML validation framework; implementation: final notebook, Experiment 1.

How do KNN and Decision Trees classify minerals?

Both receive the same pixel features, but learn and predict in different ways.

K-NEAREST NEIGHBORS

FIT Store labeled training-image vectors.

PREDICT Compare pixel distances and find the nearest examples.

NEW IMAGE
67,500 values

→

k NEIGHBORS
k = 5

→

CLASS VOTE
Quartz

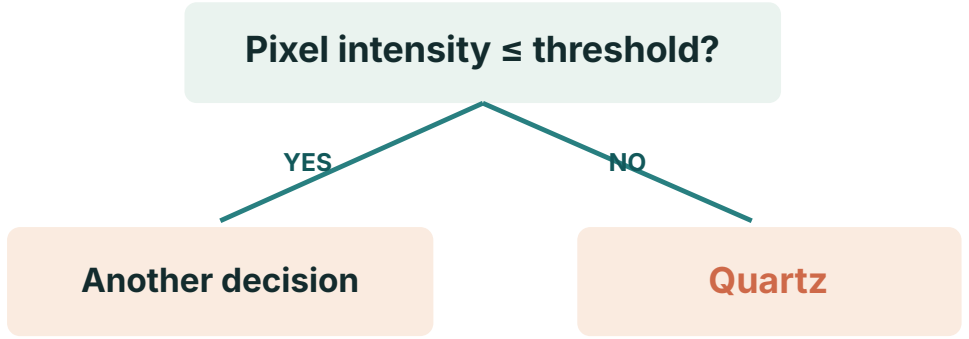
Example neighbors:

Q Q Q B P → Quartz

DECISION TREE

FIT Learn pixel thresholds that reduce Gini impurity.

PREDICT Follow learned branches to a mineral-label leaf.



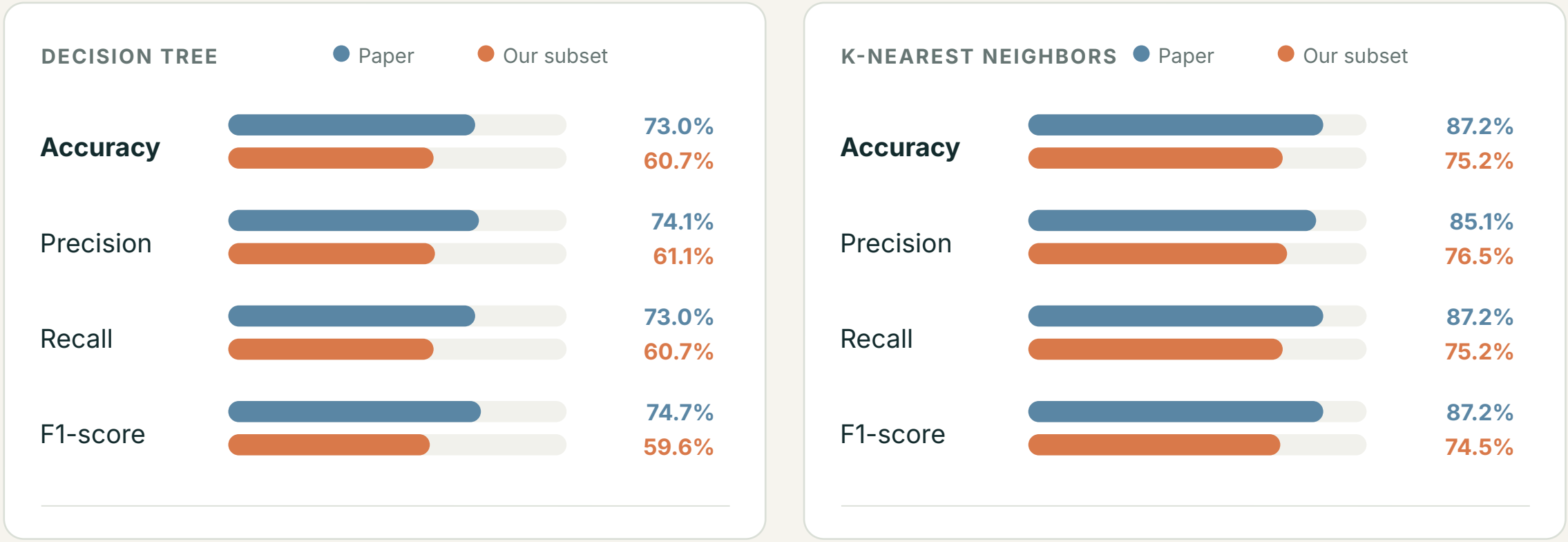
Simplified example — not a fitted tree.

KNN keeps training examples; Decision Tree keeps learned rules.

Conceptual diagrams. Methods: final notebook, Baseline Experiment 1.

Published vs reproduced: all four metrics

150 × 150 test images · Decision Tree and KNN evaluated on different available datasets.



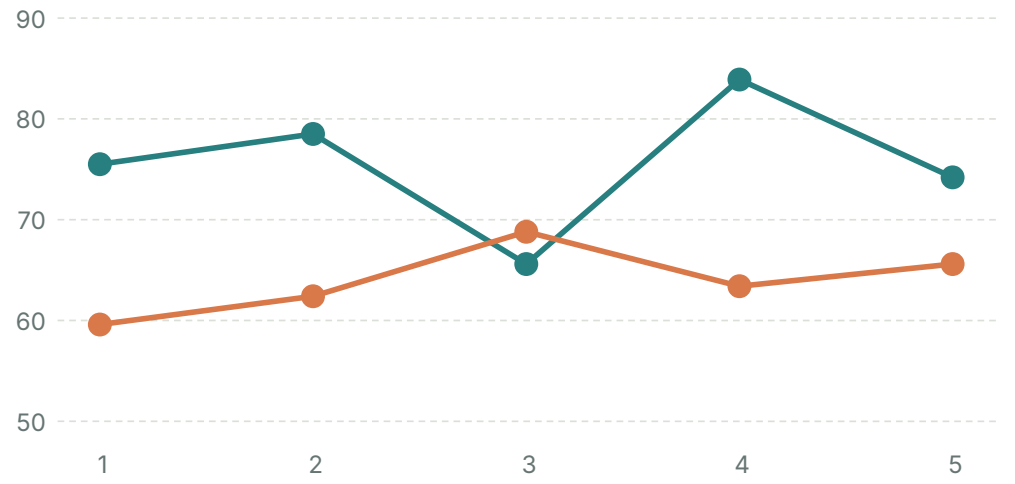
Paper: 2,500 documented images | Our evaluation: 583 available images (117 held out)

Paper: Amer et al. (2025), Table 3. Our metrics: saved notebook Experiment 1 (weighted averages).

Cross-validation: results depend on the fold

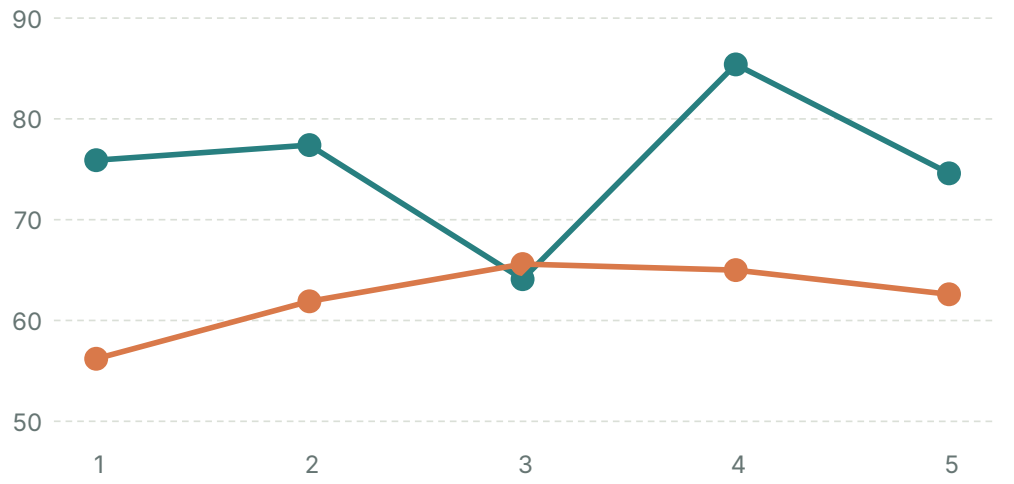
Same model, different validation images · five stratified folds of the 466 training samples.

ACCURACY (%)



KNN 75.5 ± 6.7 DT 64.0 ± 3.5

MACRO-F1 (%)



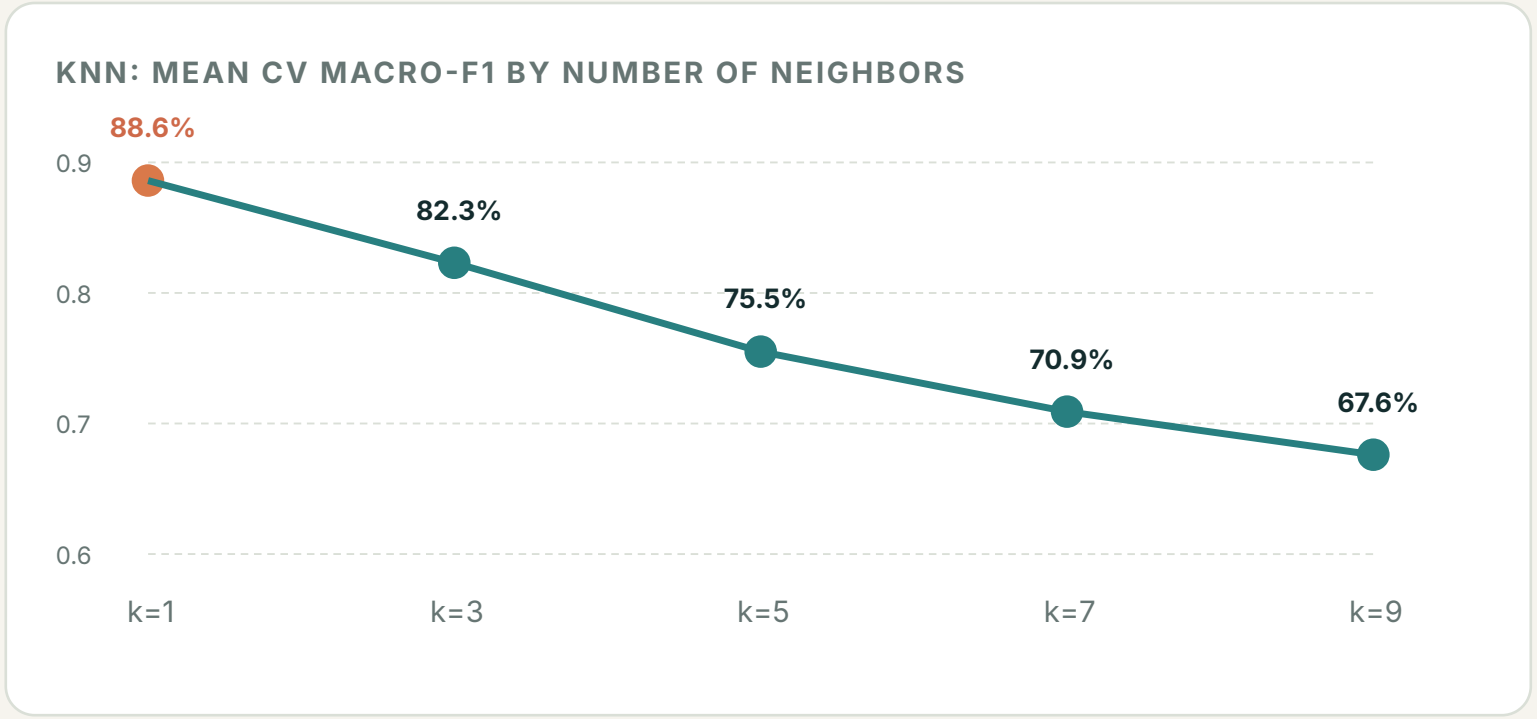
KNN 75.5 ± 7.6 DT 62.3 ± 3.7

KNN Decision Tree

Mean ± standard deviation in percentage points; 117-image holdout excluded from CV.

One nearest neighbor changes the result

GridSearchCV selected hyperparameters using mean macro-F1 across five folds.



k = 1

Best KNN setting

Decision Tree

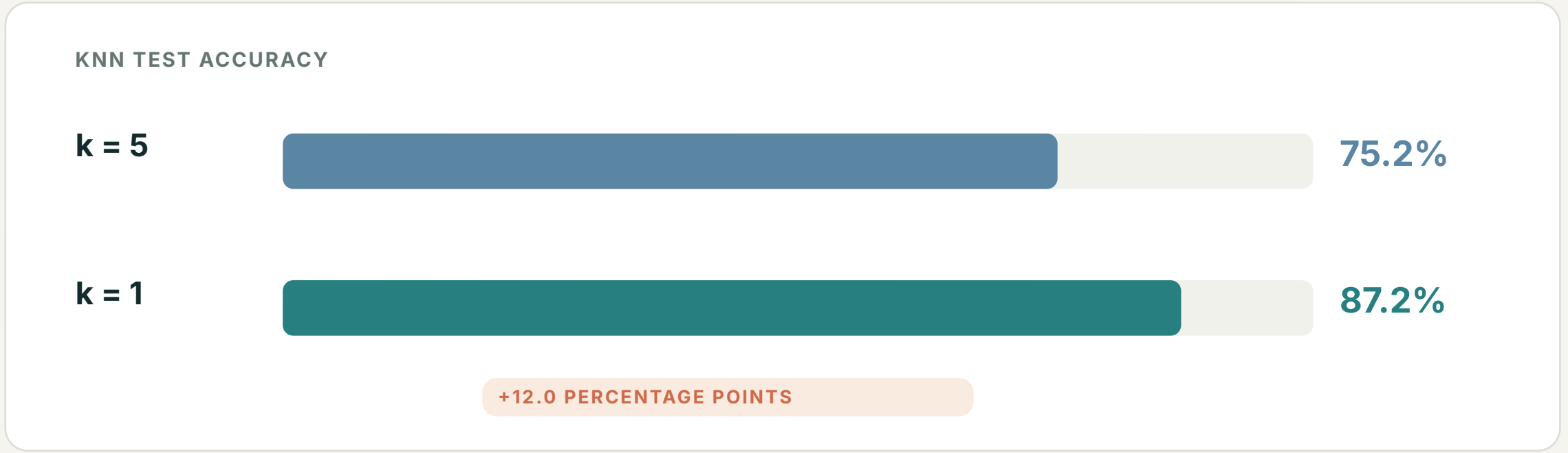
Best depth: 10

No holdout gain for DT

Search: KNN k={1,3,5,7,9}; DT max_depth={5,10,15,None}. Selection metric: mean macro-F1.

The tuned model gains 12 percentage points

On the same 117-image holdout, k=1 outperformed the baseline k=5.



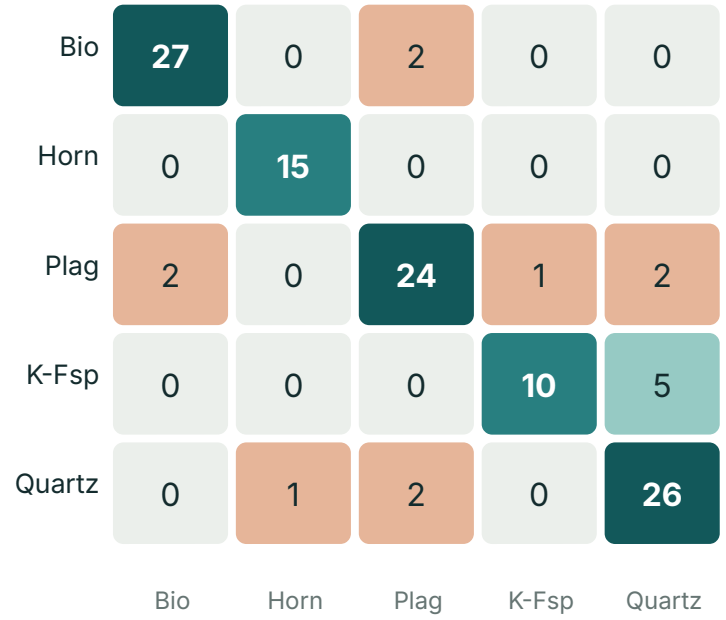
Caution: the same holdout was evaluated repeatedly; this is an observed split-specific improvement.

Final evaluation on 117 images; tuned test macro-F1 for KNN = 0.870.

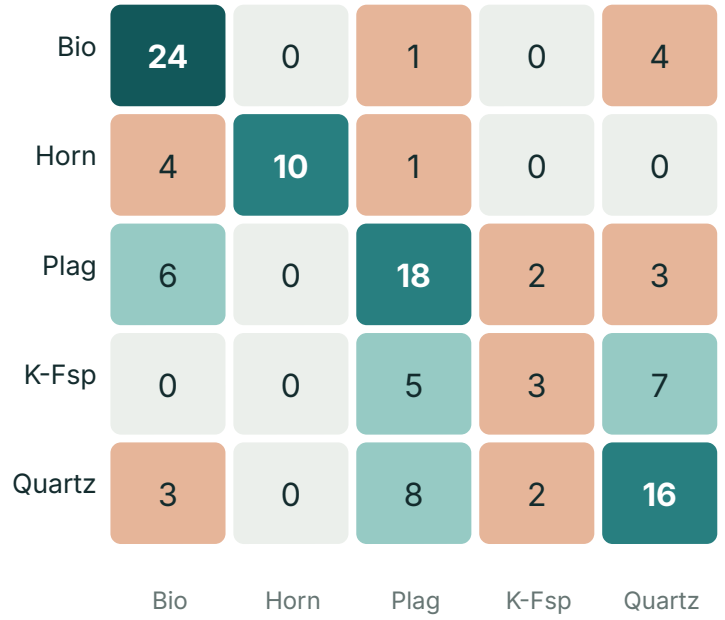
Where do the models make mistakes?

Confusion matrices show that Potassium feldspar is especially challenging.

KNN · 102 / 117 CORRECT



DECISION TREE · 71 / 117 CORRECT

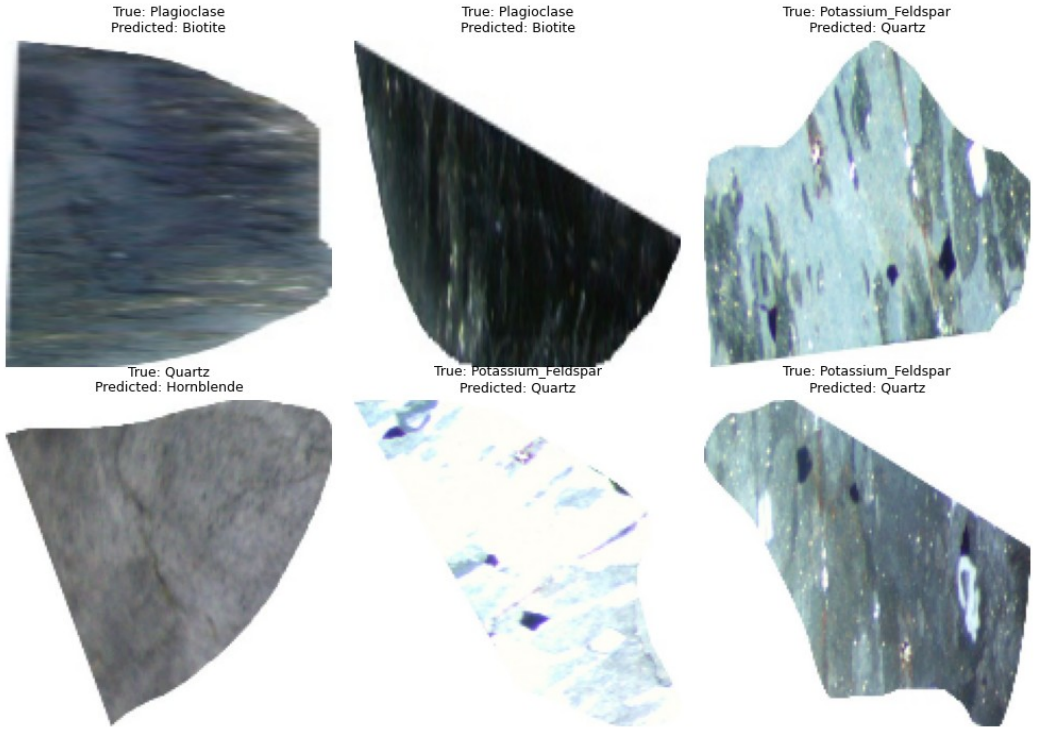


KNN: 5 of 15 Potassium-feldspar photos were misclassified as Quartz.

Rows = true labels; columns = predictions. Notebook, Experiment 4; tuned 150×150 models.

The errors are visible in real images

Six examples of test images classified incorrectly by tuned KNN.



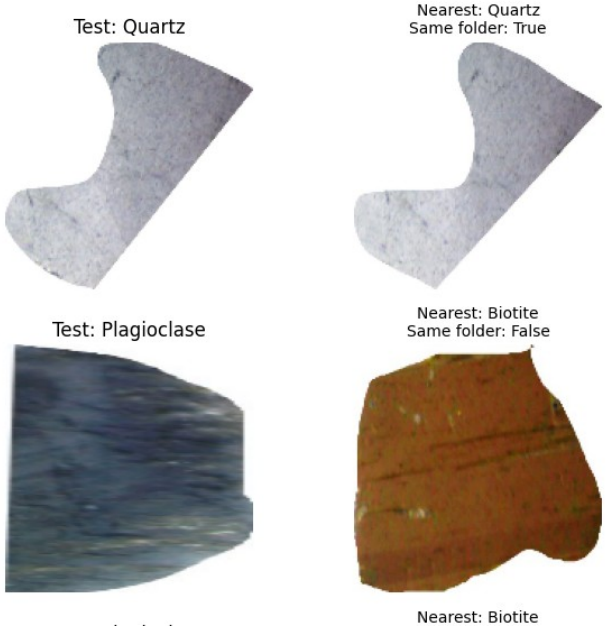
What we can say

- Several K-feldspar images were predicted as Quartz.
- Appearance changes across rotation angles.
- Six examples cannot prove which visual cues caused each error.

Source: saved Colab outputs, Experiment 4; examples are actual test images, not stock photography.

What does k = 1 really match?

Nearest-neighbor provenance links high accuracy to shared acquisition folders.



NEAREST NEIGHBOR FROM THE SAME FOLDER

101 / 117

101 / 101 correct

when the nearest neighbor shares the folder

Only 1 / 16 correct otherwise

A same-folder match guarantees the right label here because each folder contains just one mineral class.

Experiment 4, Euclidean nearest neighbors on 150×150 flattened RGB.

A stronger model — but the same evaluation unit

We tested ImageNet-pretrained EfficientNet-B0 and ConvNeXt-Tiny using transfer learning.

CNN DATA SPLIT

396	70	117
train	validation	test



Same 117 test images as classical ML

TWO-STAGE TRANSFER LEARNING

1

Freeze backbone

Train a 5-class head

→

2

Fine-tune

Last backbone stage

ImageNet pretrained · input 224×224 · augmentation on train only

Experiment 5 in final Colab notebook: EfficientNet-B0, ConvNeXt-Tiny, validation-based checkpoint selection.

CNNs improve every reported test metric

Results for the 117-image holdout · Precision, Recall, F1 are weighted averages.

MODEL & TRAINING STAGE		ACCURACY	PRECISION	RECALL	F1 (W)	MACRO-F1
EfficientNet-B0	Head only	**96.6%**	97.0%	96.6%	96.6%	96.6%
EfficientNet-B0	Fine-tuned	**99.1%**	99.2%	99.1%	99.1%	99.3%
ConvNeXt-Tiny	Head only	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**
ConvNeXt-Tiny	Fine-tuned	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**

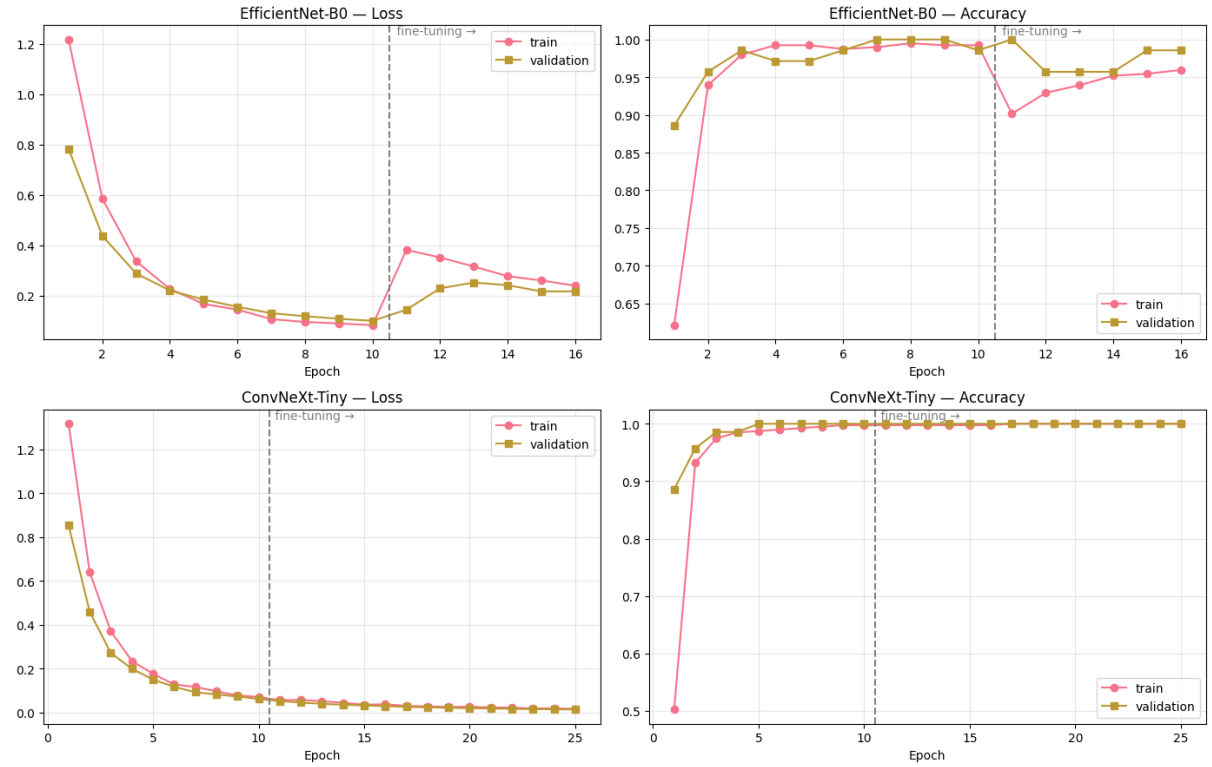
ConvNeXt-Tiny was perfect on this test set even with a frozen pretrained backbone.

Important: 100% on images from known folders ≠ verified 100% on independent crystals.

Source: final notebook, Experiment 5; 396 train / 70 val / 117 test; transfer learning from ImageNet.

Learning curves: rapid convergence on this split

Train and validation traces illustrate why high accuracy alone is insufficient evidence.



WHAT THE CURVES SHOW

EfficientNet-B0

Validation accuracy reaches 100%, but fluctuates after fine-tuning.

ConvNeXt-Tiny

Validation accuracy remains at 100% over many epochs.

Original plots from Experiment 5, training/validation learning curves; dotted line marks fine-tuning.

The unit of splitting is the real issue

The test set contains unseen photographs—not necessarily unseen crystals.

ILLUSTRATIVE IMAGE-LEVEL SPLIT OF ONE ACQUISITION FOLDER

ANGLE 0°	ANGLE 5°	ANGLE 10°	ANGLE 15°	ANGLE 20°	ANGLE 25°
TRAIN	TRAIN	TEST	TRAIN	TRAIN	TEST

Same folder / potentially same physical crystal

8 / 8 groups

appear in both train and test

117 / 117

test images belong to groups already seen during training

Group overlap: final notebook Dataset Audit (117/117). Physical crystal identity remains unconfirmed.

Can someone else repeat the experiment?

We document what can be rerun—and what the available data do not allow us to verify.

REPRODUCIBLE EXECUTION

01 Google Colab

Install, download the archive, run cells top-to-bottom

02 Fixed methodology

Seed 42 · 80/20 holdout · stratified 5-fold CV

03 Recorded outputs

Metrics, plots, confusion matrices and model files

INCLUDED WITH THE PROJECT

Notebook	Five fully documented experiments
README	Execution steps and data limitations
Requirements	Pinned package versions and environment notes

Limits: Colab-focused paths, external pretrained weights, and no verified crystal-disjoint test set.

Files: final Colab notebook, README.md, requirements.txt; data loaded from the public experimental ZIP.

CONCLUSION

A benchmark is only as strong as its test-set independence.

01

Reproduced

DT and KNN on available data

02

Extended

CV, tuning, error analysis, CNN

03

Identified

Data–metadata mismatch and group overlap

100% accuracy on 117 related images ≠ proven accuracy on unseen crystals.

Amer et al. (2025) · Scientific Data · DOI: 10.1038/s41597-025-05879-9